

BISWAJIT MANDAL

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Software Engineer · Python & C++ · Backend APIs · Machine Learning · Distributed Systems

SUMMARY

Second-year Computer Engineering student at Thapar University, seeking a Software Development role. Proficient in Python, C++, and JavaScript with practical experience in backend development, REST APIs, object-oriented design, distributed systems, databases (SQL and NoSQL), automated testing, and Agile/Scrum — applied across two independently built, production-deployed AI applications.

EDUCATION

Thapar Institute of Engineering & Technology — Patiala, Punjab

Aug 2024 – Jun 2028

B.Tech, Computer Engineering

Coursework: Algorithms & Data Structures, OOP, Database Systems, Distributed Systems, Operating Systems

SKILLS

Languages: Python, C++, C, JavaScript

Frameworks & Tools: FastAPI, TensorFlow/Keras, scikit-learn, pandas, NumPy, Docker, GitHub Actions, AWS, Linux

Databases: PostgreSQL, MySQL (relational) | MongoDB, Redis (NoSQL)

Practices: Object-oriented design, REST API design, automated testing (unit/integration/smoke), CI/CD, code reviews

PROJECTS

Neural Predictive Maintenance System — Smart Factory AI

Apr – May 2026

Python · TensorFlow/Keras · FastAPI · scikit-learn · Docker Compose · PostgreSQL · Redis · MQTT · GitHub Actions

- **Designed and shipped a real-time ML system** that ingests 9 industrial sensor streams (temperature, torque, RPM, vibration, pressure, acoustic, humidity) and predicts machine failure probability — production-ready with ROC-AUC and PR-AUC evaluation.
- **Built a neural temporal sensor-fusion model** using 3 parallel Conv1D branches (thermal, mechanical, electrical) in Keras, combined with a leakage-free Random Forest baseline using physics-based engineered features.
- **Developed 6 RESTful FastAPI endpoints** for live inference, stream simulation, health checks, and diagnostics — with input validation, structured logging, and security compliance built in.
- **Implemented full automated test coverage** (unit, integration, smoke) across feature engineering, model training, streaming, and API layers — enforced via GitHub Actions CI/CD across test and production environments.
- **Containerized the entire distributed system** using Docker Compose with 8 services (API, dashboard, PostgreSQL, Redis, MQTT broker, replay, bridge, monitoring) — zero-downtime, reproducible deployments.

Hospital Drug Shortage Simulation — Open-Source RL Benchmark

Mar – Apr 2026

Python · Reinforcement Learning · Hugging Face Spaces · FastAPI · Distributed Simulation · Automated Evaluation

- **Architected a distributed multi-agent RL environment** spanning 6 task families and 3 difficulty tiers — evaluating AI decision-making under budget pressure, supply-chain risk, clinical constraints, and perishable inventory management.
- **Implemented standardized action interfaces, deterministic graders, and reproducible task distributions** following common-component design principles; added ablation testing to quantify the impact of 3+ key agent capabilities.
- **Deployed live on Hugging Face Spaces** with a validator-safe RESTful API contract — publicly accessible, used by the AI research community for reproducible benchmarking across distributed environments.
- **Delivered 4 task categories using Agile sprints and user stories**, with full documentation, integration tests, and reproducible evaluation pipelines from day one.

ACHIEVEMENTS

- **Shipped 2 production-grade AI systems as a second-year student** — both fully containerized, tested, and publicly live; built entirely independently outside coursework.
- **Graduate-level project depth in year two** — neural sensor fusion, RL benchmarking, and distributed microservices are topics typically covered in final-year or postgrad programs.
- **Certified: Design & Analysis of Algorithms (CodeChef, May 2026)** — complexity analysis, dynamic programming, graph algorithms, and optimization.